

46 YEARS OF EXCELLENCE | NARAYANA IIT - JEE ACADEMY - INDIA

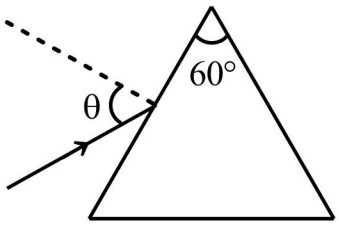
Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **Date: 05.01.2025**
Time: 3HRS **SPCL_GTM-01** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. If refractive index of glass prism is $\sqrt{2}$, then angle of incidence θ required for the minimum deviation through prism is



A) 37^0 B) 60^0 C) 45^0 D) 53^0

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2. A light, inextensible uniform bar of mass M and length L is tied to a string of length L as shown in the figure. The bar turns in a horizontal circle with angular speed 'ω' so that one end of string remains fixed.

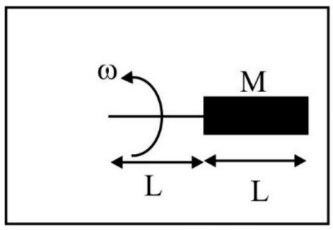


Table
(Top - view)

The tension in the string is

A) $m\omega^2 L$ B) $2m\omega^2 L$ C) $3m\omega^2 \frac{L}{2}$ D) $m\omega^2 \frac{L}{2}$

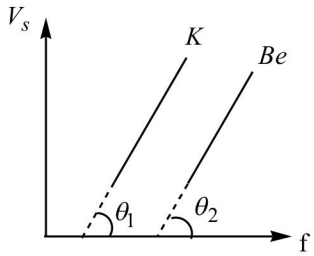
3. A particle moves along the curve $y^2 = 2x$ in the first quadrant of co-ordinate axes. The x coordinate (in meter) of particle depends on time t (in seconds) as $x = \frac{t^2}{2}$. The acceleration of the particle at $t = 2\text{s}$ is____

A) $\left(1\text{m/s}^2\right)\hat{i}$ B) $\left(0.5\text{m/s}^2\right)\hat{j}$ C) $\left(1\text{m/s}^2\right)\left(\hat{i}-\hat{j}\right)$ D) $\left(2\text{m/s}^2\right)\hat{i}$

4. Bulk modulus of an ideal gas for a process $P = \frac{\alpha}{T^2}$ is (where α is constant)

A) $\frac{3}{2}P$ B) $\frac{2}{3}P$ C) P D) $\frac{1}{2}P$

5. Figure shows the graph when the stopping potential V_s is plotted against the frequency of incident radiations for two metals Potassium (K) and Beryllium (Be). Select the correct alternative



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A) $\theta_1 > \theta_2$

B) $\theta_1 < \theta_2$

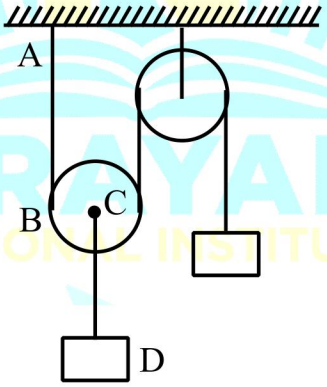
C) The threshold wavelength of K is larger

D) The threshold wavelength of Be is larger

6. A uniform magnetic field exists in region given by $\vec{B} = \left(3\hat{i} + 4\hat{j} + 5\hat{k}\right)\text{T}$. A rod of length 5m placed along y-axis is moved along x-axis with constant speed 1m/sec. Then induced emf in the rod will be

A) zero B) 25 V C) 20 V D) 15 V

7. Both the strings, show in figure, are made of same material and have same cross section. The pulleys are light. The wave speed of a transverse wave in the string AB is v_1 and in CD it is v_2. The v_1 / v_2 is



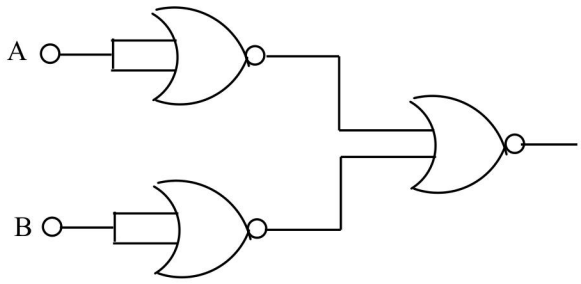
A) 1 B) 2 C) $\sqrt{2}$ D) $1/\sqrt{2}$

8. For a monoatomic ideal gas undergoing an adiabatic change, the relation between temperature and volume $TV^x = \text{constant}$, where x is

A) $\frac{7}{5}$ B) $\frac{2}{5}$ C) $\frac{2}{3}$ D) $\frac{1}{3}$

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9. The combination shown in figure acts like



A) AND gate B) OR gate C) NAND gate D) XOR gate

10. Which of the following is correct?

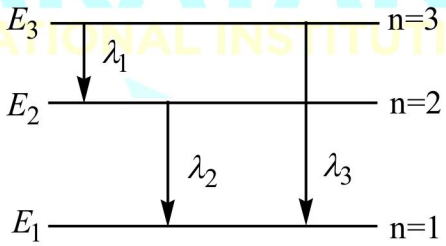
A) A time varying magnetic field produces electric field

B) A time varying electric field produces magnetic field

C) If the instantaneous induced electric field is E and at the same time magnetic field is B then E=Bc, where c= speed of light in the medium

D) All of these

11. Three energy levels of an atom are shown in the figure. The wavelength corresponding to three possible transitions are λ_1, λ_2 *and* λ_3. Choose the correct relation



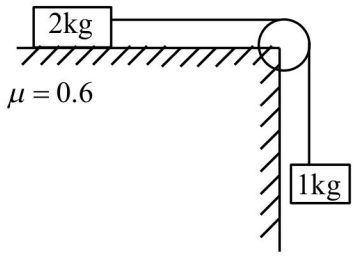
A) $\lambda_1 + \lambda_2 = \lambda_3$ B) $\frac{1}{\lambda_1} + \frac{1}{\lambda_2} = \frac{1}{\lambda_3}$ C) $\lambda_1\lambda_2 = \lambda_3^2$ D) $\lambda_1\lambda_2 + \lambda_2\lambda_3 = \lambda_1\lambda_3$

12. A very large number of particles of same mass m are kept at horizontal distances of 1m, 3m, 4m, 8m …(upto infinity) from origin. The total gravitational potential at origin is

A) $-8Gm$ B) $-3Gm$ C) $-4Gm$ D) $-2Gm$

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13. In the arrangement shown, string is light and inextensible. There is no friction at the pulley. The tension in the string is

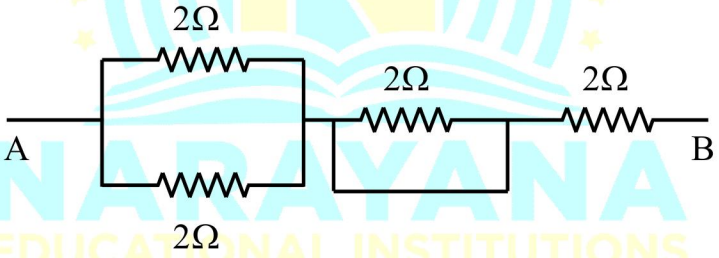


A) 10N B) 8N C) 12N D) Zero

14. The pressure inside two soap bubbles are 1.01 Atm and 1.02 Atm. The ratio of their respective volume is

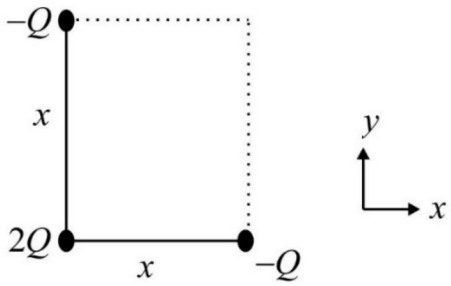
A) 8 B) 4 C) 16 D) $\frac{1}{8}$

15. Find equivalent resistance between A and B



A) 5Ω B) 3Ω C) 2Ω D) 8Ω

16. Consider three point charges placed on the vertices of a square (of side x) as shown. Electric dipole moment of the system is



A) $-Qx\left(\hat{i}+\hat{j}\right)$ B) $\sqrt{2}Qx\left(\hat{i}+\hat{j}\right)$

C) $\left(\sqrt{2}-1\right)Qx\left(\hat{i}-\hat{j}\right)$ D) $2Qx\hat{i}$

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17. The dimensional formula for

$$\frac{(\text{Capacitance})(\text{Electrostatic Potential})^2}{(\text{Permitivity})(\text{Electric field})^2}$$

is same as that of:

A) Mass B) Density C) $\frac{\text{Mass}}{\text{Density}}$ D) $\frac{\text{Density}}{\text{Mass}}$

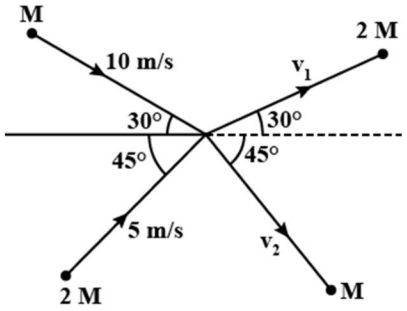
18. A long cylindrical wire of radius R carries a current I distributed uniformly over its cross-section. Find the maximum magnetic field produced by this wire

A) $\frac{\mu_0 I}{2\pi R}$ B) $\frac{8\mu_0 I}{\pi R}$ C) $\frac{\mu_0 I}{4\pi R}$ D) $\frac{2\mu_0 I}{\pi R}$

19. A particle is projected at time $t = 0\text{s}$ with speed v at angle θ with horizontal. The particle is at same height h ($h <$ maximum height) at two different times $t = t_1$ and $t = t_2$. If the value of $t_1 \times t_2$ is $\frac{xh}{g}$, then the value of x is _____.

A) 4 B) 3 C) 2 D) 1

20. Two particles, of masses M and 2M, moving, as shown, with speeds of 10 m/s and 5 m/s, collide elastically at the origin. After the collision, they move along the indicated directions with speeds v_1 and v_2 respectively. The value of v_1 is nearly:



A) 2.5 m/s B) 3.9 m/s

C) 6.5 m/s D) 9.2 m/s

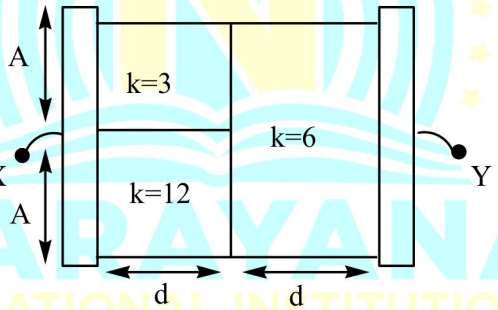
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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

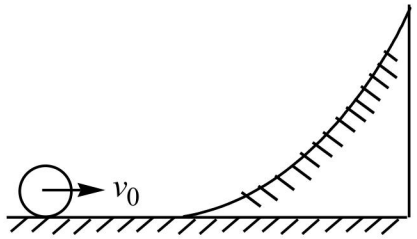
21. The reading of spring balance when a block is suspended from it in air, is 60 N. This reading is changed to 40 N when the block is immersed in water. Find the specific gravity of the block.

22. Space between plates of parallel plate capacitor is filled with three dielectric slabs as shown. The different values of dielectric constants (k) are shown in the diagram for the three dielectrics. Equivalent dielectric constant for the arrangement shown (between points X and Y) is



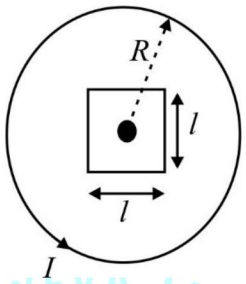
23. Two waves having the intensities in the ratio of 16 : 9 produce interference. The ratio of maximum to minimum intensity is equal to x:1. Then the value of x is _____

24. A solid sphere is rolling on a horizontal surface with velocity v_0. It rolls up without slipping on a curved incline as shown. The maximum height gained by centre of sphere is $\frac{nv_0^2}{10g}$. Then the value of n is_____.



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25. The diagram shows two co-planar conducting loops whose centres coincide. If $l << R$, then the mutual inductance is $\frac{\mu_0 l^2}{NR}$, then the value of N is___.

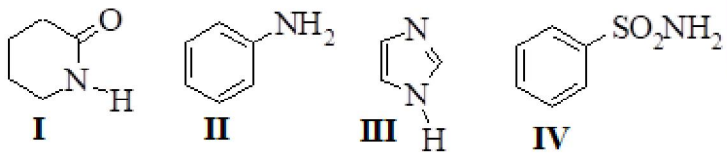


CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Among the following compounds, which is the **most basic**, which has the **most acidic** hydrogen respectively?



A) II = strongest base, I = strongest acid

B) IV = strongest base, III = strongest acid

C) III = strongest base, IV = strongest acid

D) II = strongest base, III = strongest acid

27. The ratio of time periods taken by electrons in 1st and 3rd orbits of He^+ ion, for each revolution is

A) 1 : 9 B) 1 : 27 C) 8 : 27 D) 8 : 9

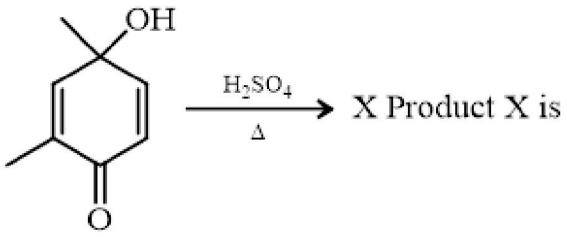
28. The equilibrium constant of mutarotation of $\alpha - D$-glucose to $\beta - D$-glucose is 1.8. If specific rotation of $\alpha - D$-glucose is 112^0 and that of mixture is 59^0. The specific rotation of $\beta - D$-glucose is_____

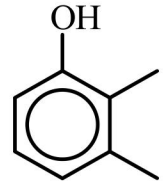
A) -112^0 B) -59^0 C) $+29.6^0$ D) -29.6^0

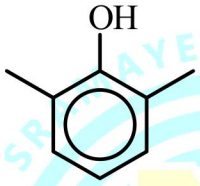
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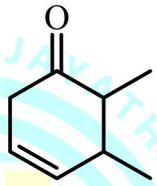
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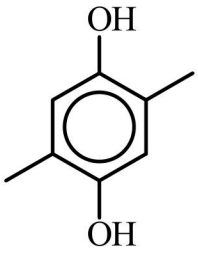
29.



A) 

B) 

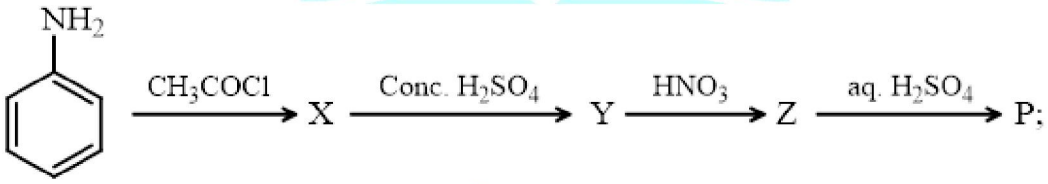
C) 

D) 

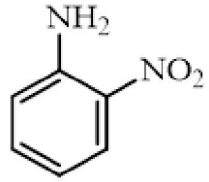
30. Which of the following compound cannot give Fehling's test?

A) Benzaldehyde B) Methanal C) Propionaldehyde D) Valeraldehyde

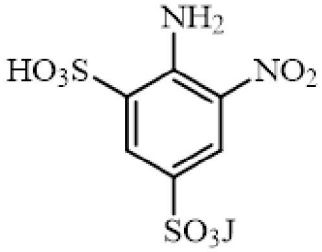
31.

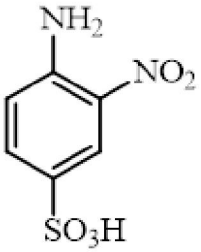


The product 'Z' of the above reaction is_____

A) 

B) 

C) 

D) 

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32. Which of the following is tetrahedral and paramagnetic complex?

A) $[NiCl_4]^{2-}$ B) $[Ni(CN)_4]^{2-}$ C) $[Cu(NH_3)_4]^{2+}$ D) $[Ni(CO)_4]$

33. The oxidation states of bromine in Br_3O_8 are_____

A) $+1,+3$ B) $+5,+7$ C) $+4,+6$ D) $+5,+3$

34. 2 moles of nitrogen gas is expanded at a constant pressure, in this process the ratio of internal energy change to the enthalpy change is____

A) 0.60 B) 0.75 C) 0.71 D) 0.50

35. Which of the following is not a redox reaction ?

A) $2KClO_3 + I_2 \rightarrow 2KIO_3 + Cl_2$

B) $5NaBr + NaBrO_3 + 6HCl \rightarrow 6NaCl + 3Br_2 + 3H_2O$

C) $CaOCl_2 + 2NaI + 2HCl \rightarrow I_2 + CaCl_2 + H_2O + NaCl$

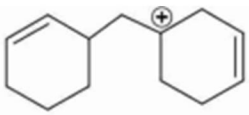
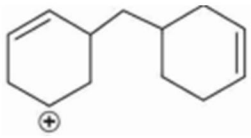
D) $NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$

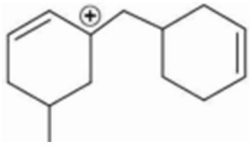
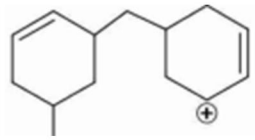
36. Which of the following is optically active?

A) 2 – propanol B) β – hydroxybutyraldehyde

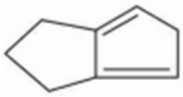
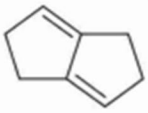
C) p – xylene D) β – methylbutyraldehyde

37. Which of the following is the most stable carbocation?

A)  B) 

C)  D) 

38. Which of the diene has highest heat of hydrogenation?

A)  B)  C)  D) 

For More Material Join: @JEEAdvanced_2025

39. A liquid is placed in a closed vessel of volume V lit. If the volume of the vessel is reduced to half by keeping the temperature constant, the change in vapour pressure of the liquid is___

A) doubled to its initial value B) half to its initial value

C) no change in its value

D) vapour pressure is an extensive property

40. For a reaction, $3A + B \longrightarrow 2C$

$-\frac{d[A]}{dt} = K[A][B]$, the expression for $\frac{d[C]}{dt}$ will be

A) $\frac{2}{3}K[A][B]$ B) $\frac{3}{2}K[A][B]$ C) $\frac{1}{3}K[A][B]$ D) $\frac{1}{2}K[A][B]$

41. For the reaction $SrCO_3(s) \rightleftarrows SrO(s) + CO_2(g)$, at equilibrium the pressure of $CO_2(g)$ depends on

A) Amount of $SrCO_3(s)$

B) Amount of $SrO(s)$

C) Amounts of both $SrCO_3(s)$ and $SrO(s)$

D) Temperature of the system

42. On increasing the temperature, the ionic product of water increases. $\left[H_2O \rightleftharpoons H^+ + OH^-, K_w = 10^{-14}\ at\ 25^0 C\right]$ Which of the following statement is correct regarding dissociation of water?

A) water becomes basic on decreasing the temperature

B) pH of water increase on increasing the temperature

C) water will be neutral irrespective of temperature change

D) pH cannot be predicted on given information

For More Material Join: @JEEAdvanced_2025

43. What is the pH of the acid used in the given half cell reaction

$Pt|Cr_2O_7^{2-}(0.01M), H^+(pH=X)|Cr^{3+}(0.1M), H_2O_{(l)}$

Given $E^0_{Cr_2O_7^{2-},H^+/Cr^{3+}} = 1.33V$ and $E_{cell} = 1.05V$

A) 14 B) 7 C) 2 D) 0

44. Consider the following statements about lactose

1. Lactose on hydrolysis yields equimolar mixture of D-glucose and D-galactose
2. Lactose is a poly-saccharide
3. Lactose is a reducing sugar
4. In lactose, the glycosidic linkage is between C-1 of galactose and C-4 glucose unit.

Which of the statements are correct?

A) 1 and 3 only B) 1 and 4 only C) 2 and 3only D) 1, 3 and 4 only

45. Dibutylamine, $(C_4H_9)_2NH$, and anisole, $C_6H_5OCH_3$, have similar boiling points, and are relatively insoluble in water. How might a mixture of these compounds be separated into the pure components?

a) **i)** dissolve mixture in diethylether; **ii)** extract the anisole into 10% aqueous NaOH

b) **i)** dissolve mixture in diethylether; **ii)** extract the amine into 10% aqueous HCl

c) **i)** dissolve mixture in diethylether; **ii)** extract the amine into 10% aqueous NaOH

d) **i)** treat the mixture with $PhSO_2Cl$; **ii)** hydrolyse the precipitate formed in to amine using aq. NaOH

A) (a) and (b) only B) (b), (c) and (d) only

C) (b) and (d) only D) (a), (b), (c) and (d)

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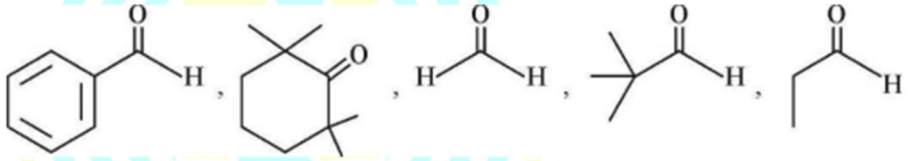
SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. $C_2H_5OH + KMnO_4 \rightarrow C_2H_4O_2 + MnO_2 + KOH + H_2O$ is the unbalanced equation. If 5g of ethanol and excess amount of aqueous $KMnO_4$ are reacted and 5.5g of $C_2H_4O_2$ is obtained. What is the percent yield of the reaction?

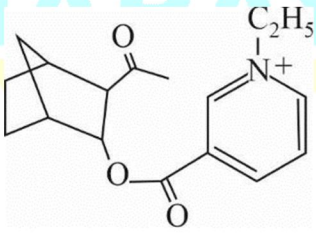
[Given: Atomic mass of H= 1, C = 12, O = 16]

47. Of the following, how many compounds can undergo aldol reaction?



48. How many of the following species are polar? $CO_2, BF_3, XeF_2, ClF_3, NH_3, XeF_4, H_2O$

49. The number of chiral centres present in the following compound are_____.



50. An ideal solution is formed by mixing of 460 g. of ethanol with "X"g of methanol. The total vapour pressure of the solution is 72 mm of Hg. The vapour pressure of pure ethanol and pure methanol are 48 and 80 mm of Hg respectively. The value of "X" is_____

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MATHS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. $2x+y=0$ is the equation of diameter of a circle which touch the lines $4x-3y+10=0$ and $8x-6y-60=0$. The square of the length of intercept made by the circle on $x-$ axis is___

A) 48 B) 38 C) 28 D) 18

52. A box contains 15 green and 10 yellow balls. If 10 balls are drawn randomly, one by one with replacement. The variance of the number of green balls drawn is:

A) $\frac{6}{25}$ B) $\frac{12}{5}$ C) 6 D) 4

53. If $0<c<a<b<d$, then roots of the equation $bx^2+\left(1-b\left(c+d\right)\right)x+bcd-a=0$ must be___

A) Real and distinct and one lies between c and a

B) Real and distinct and one lies between a and b

C) Real and distinct and one lies between c and d

D) Not real

54. The vertices of a parallelogram $ABCD$ are $A(\alpha,1)$, $B(13,6)$, $C(13,\beta)$ and $D(3,16)$. If the line $\alpha x+\beta y=k$ divides the parallelogram into two congruent parts, then $k=$

A) 185 B) 205 C) 225 D) 255

55. The exhaustive set of solution of the inequality $\frac{\left(e^x-1\right)\left(2x-3\right)\left(x^2+x+2\right)}{\left(\sin x-2\right)\left(x\right)\left(x+1\right)}\le 0$ is_____

A) $\left[\frac{3}{2},\infty\right)$ B) $\left(-\infty,-1\right)\cup\left[\frac{3}{2},\infty\right)$

C) $\left(-1,0\right)\cup\left[\frac{3}{2},\infty\right)$ D) $R-\{0,-1\}$

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56. If $f(x)=-1+|x-2|, 0\le x\le 4$ and $g(x)=fof(x)$, then $gogogog(1)=$

A) 1 B) 101 C) 1995 D) 2025

57. The system of equations $x-2y+z=a$, $2x+y-2z=b$ and $x+3y-3z=c$ have atleast one solution, then

A) $a+b+c=0$ B) $a-b+c=0$ C) $b-a+c=0$ D) $c-a-b=0$

58. Let 'a' be the sum of the coefficients in the expansion of $\left(\frac{1}{4}-15x+15x^3\right)^{2025}\left(17x-17x^5+4\right)^{2026}$ and 'b' be the value of $\underset{x\to-\pi}{Lt}\frac{\int_0^{\sin x}\sin^{-1}t\,dt}{(x+\pi)^2}$. If the equations $x^3+x^2-ax-8b=0$ and $x^2+px+2p=0$, $p\in R$ have two common roots, then the value of p is:

A) 1 B) 2 C) –1 D) –2

59. $\underset{x\to0}{Lt}\frac{x\sin(\sin x)-\sin^2 x}{x^6}$ is equal to:

A) $\frac{1}{6}$ B) $\frac{1}{12}$ C) $\frac{1}{18}$ D) $\frac{1}{24}$

60. Let $\bar{a}=\hat{i}-3\hat{j}+4\hat{k}$, $\bar{b}=6\hat{i}+4\hat{j}-8\hat{k}$, $\bar{c}=5\hat{i}+2\hat{j}+5\hat{k}$ and a vector $\bar{r}$ satisfy $\bar{r}\times\bar{b}=\bar{c}\times\bar{b}$ and $\bar{r}\cdot\bar{a}=0$ then the value of $\frac{|\bar{b}|}{|\bar{r}-\bar{c}|}$ is:

A) 1 B) 2 C) 3 D) 4

61. Consider the lines $L_1:\frac{x-7}{3}=\frac{y-5}{2}=\frac{z-3}{1}$ and $L_2:\frac{x-1}{2}=\frac{y+1}{4}=\frac{z+1}{3}$. If a line $L=0$ whose direction ratios are $(2, 2, 1)$ intersects L_1 and L_2 at A and B. Then the distance from the point $(3, 3, 2)$ to the line L is:

A) 0 B) 1 C) 2 D) 3

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62. A function $y = f(x)$ satisfying the differential equation

$\frac{dy}{dx} \cdot \sin x - y \cdot \cos x + \frac{\sin^2 x}{x^2} = 0$ is such that $y \to 1$ as $x \to \frac{\pi}{4}$, then $\underset{x \to 0}{Lt} f(x) =$

A) 0 B) 1 C) 2 D) 3

63. A bag contains 5 white, 6 green, 7 blue, 8 red and 9 black balls. If 3 balls are drawn one by one with replacement after each drawing, then the probability that 3rd ball drawn is green is, if it is known that second ball is green:

A) $\frac{1}{35}$ B) $\frac{2}{35}$ C) $\frac{6}{35}$ D) $\frac{4}{35}$

64. If $f\left(\frac{x+y}{2}\right) = \frac{f(x)+f(y)}{2} \forall x, y \in R$, $f(0) = 1$, $f'(0) = -1$ and

$g(x) = \begin{cases} A + B\sin^{-1} x & ; x \geq 0 \\ f(x) & ; x < 0 \end{cases}$ is differentiable $\forall x \in R$, then the value of $A^3 + B^3$ is:

A) 10 B) 8 C) 2 D) 0

65. If α, β are the values of k for which the foci of ellipse $\frac{x^2}{k^2 a^2} + \frac{y^2}{a^2} = 1$ and hyperbola $\frac{x^2}{a^2} - \frac{y^2}{a^2} = 1$ coincide, then the value of $(\alpha + \beta)^2$ is____

A) 0 B) 1 C) 4 D) 9

66. The sum of the first 50 terms of the series

$(1^2 - 1 + 1)1! + (2^2 - 2 + 1)2! + (3^2 - 3 + 1)3! + \ldots..$ is:

A) 52! B) $1 + 49 \cdot 51!$ C) $52! - 1$ D) $50 \cdot 51! - 1$

67. If the domain of the function $f(x) = \cos^{-1}\left(\frac{2 - |x|}{4}\right) + \left(\log(3 - x)\right)^{-1}$ is $[a, b) \cup (b, c)$, then area of the region bounded by the curve $y^2 \leq 8x$ in (b, c) is:

A) $2\sqrt{6} - \frac{8}{3}$ B) $4\sqrt{6} - \frac{32}{3}$ C) $4\sqrt{6} - \frac{16}{3}$ D) $8\sqrt{6} - \frac{32}{3}$

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68. If $f(x)=\begin{vmatrix} x-2 & (x-1)^2 & x^3 \\ x-1 & x^2 & (x+1)^3 \\ x & (x+1)^2 & (x+2)^3 \end{vmatrix}$, then the value of $f(0)+f'(0)=$

A) 28 B) 18 C) 108 D) 8

69. If $\left(\sin^{-1}x\right)^2+\left(\cos^{-1}y\right)^2+\left(\sec^{-1}z\right)^2+\left(\operatorname{cosec}^{-1}w\right)^2=\frac{5\pi^2}{2}$, then maximum possible value of $x+y+z+w$ is:

A) 0 B) 1 C) 2 D) 4

70. If $24\int_0^{\frac{\pi}{4}}\frac{2\tan x\left(\cos^2 x+1\right)}{4+6\cos^2 x+\cos^6 x}dx=2\log_e\left(\frac{a}{b}\right)$, then the least possible integral value of $(a+b)$ is___ (where a,b are co − prime)

A) 68 B) 46 C) 11 D) 57

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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71. $\bar{a}=2\hat{i}+2\hat{j}-\hat{k}$, $\bar{b}=\hat{i}-2\hat{j}-\hat{k}$ and $\bar{c}$ is a non-zero vector such that $\bar{a}\cdot\bar{c}=3|\bar{c}|$ and $|\bar{b}\times\bar{c}|=\sqrt{212}$. If the angle between $\bar{b}$ and $\bar{c}$ is θ, then the value of $216\cos^2\theta$ is

_____

72. A functional equation satisfying $f(x+y)=2^x f(y)+4^y f(x)$ such that $f'(0)=\ln 2$, $I_1=\int_{f(0)}^{1+f(1)} x\,f\left(x(x-3)\right)dx$, $I_2=\int_{f(0)}^{1+f(1)} f\left(x(x-3)\right)dx$ and $I_1=K\,I_2$. The value of $10K$ is equal to_____

73. If for the complex numbers z satisfying $|z-2-2i|\le 1$, the maximum value of $|iz+2|$ is attained at $a+ib$ (where $a,b\in R$). Then $2a+3b$ is equal to

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74. If two lines $L_1 : x = 5, \frac{y}{3-\alpha} = \frac{z}{-2}$ and $L_2 : x = \alpha, \frac{y}{-1} = \frac{z}{2-\alpha}$ are coplanar, then sum of all possible values of α is equal to _____

75. For the function $f(x) = \int_0^{x^2} e^{-\frac{t}{4}}(4-t)\,dt$, let $p = sum$ of squares of the values of x, where $f(x)$ attains local maxima and q = sum of squares of the values of x, where $f(x)$ attains local minima, then the value of $p + 2q$ is equal to ____

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